Appendix

Title: Healthcare costs and carbon emissions of stage III melanoma surveillance imaging

Authors

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Appendix sections

A CHEERS Statement

B STROBE Statement

C MBS Items

D Imaging scan durations and carbon footprint

E AusLCI and ecoinvent methods

F Carbon valuations

A CHEERS Statement

Topic	No.	Item	Location where item is reported
Title			
	1	Identify the study as an economic evaluation and specify the interventions being compared.	Cover page
Abstract			
	2	Provide a structured summary that highlights context, key methods, results, and alternative analyses.	Page 1
Introduction			
Background and objectives	3	Give the context for the study, the study question, and its practical relevance for decision making in policy or practice.	Page 4
Methods			
Health economic analysis plan	4	Indicate whether a health economic analysis plan was developed and where available.	Page 6
Study population	5	Describe characteristics of the study population (such as age range, demographics, socioeconomic, or clinical characteristics).	Page 6
Setting and location	6	Provide relevant contextual information that may influence findings.	Page 6
Comparators	7	Describe the interventions or strategies being compared and why chosen.	Page 7
Perspective	8	State the perspective(s) adopted by the study and why chosen.	Page 8
Time horizon	9	State the time horizon for the study and why appropriate.	NA
Discount rate	10	Report the discount rate(s) and reason chosen.	NA
Selection of outcomes	11	Describe what outcomes were used as the measure(s) of benefit(s) and harm(s).	NA
Measurement of outcomes	12	Describe how outcomes used to capture benefit(s) and harm(s) were measured.	NA
Valuation of outcomes	13	Describe the population and methods used to measure and value outcomes.	NA

Topic	No.	Item	Location where item is reported
Measurement and valuation of resources and costs	14	Describe how costs were valued.	Page 8
Currency, price date, and conversion	15	Report the dates of the estimated resource quantities and unit costs, plus the currency and year of conversion.	Page 8
Rationale and description of model	16	If modelling is used, describe in detail and why used. Report if the model is publicly available and where it can be accessed.	NA
Analytics and assumptions	17	Describe any methods for analysing or statistically transforming data, any extrapolation methods, and approaches for validating any model used.	Page 9
Characterising heterogeneity	18	Describe any methods used for estimating how the results of the study vary for subgroups.	NA
Characterising distributional effects	19	Describe how impacts are distributed across different individuals or adjustments made to reflect priority populations.	NA
Characterising uncertainty	20	Describe methods to characterise any sources of uncertainty in the analysis.	NA
Approach to engagement with patients and others affected by the study	21	Describe any approaches to engage patients or service recipients, the general public, communities, or stakeholders (such as clinicians or payers) in the design of the study.	NA
Results			
Study parameters	22	Report all analytic inputs (such as values, ranges, references) including uncertainty or distributional assumptions.	Appendix C
Summary of main results	23	Report the mean values for the main categories of costs and outcomes of interest and summarise them in the most appropriate overall measure.	Page 10
Effect of uncertainty	24	Describe how uncertainty about analytic judgments, inputs, or projections affect findings. Report the effect of choice of discount rate and time horizon, if applicable.	NA
Effect of engagement with patients and others affected by the study	25	Report on any difference patient/service recipient, general public, community, or stakeholder involvement made to the approach or findings of the study	NA
Discussion			

Topic	No.	Item	Location where item is reported
Study findings, limitations, generalisability, and current knowledge	26	Report key findings, limitations, ethical or equity considerations not captured, and how these could affect patients, policy, or practice.	Page 12
Other relevant information			
Source of funding	27	Describe how the study was funded and any role of the funder in the identification, design, conduct, and reporting of the analysis	Title page
Conflicts of interest	28	Report authors conflicts of interest according to journal or International Committee of Medical Journal Editors requirements.	Title page

From: Husereau D, Drummond M, Augustovski F, et al. Consolidated Health Economic Evaluation Reporting Standards 2022 (CHEERS 2022) Explanation and Elaboration: A Report of the ISPOR CHEERS II Good Practices Task Force. Value Health 2022;25. doi:10.1016/j.jval.2021.10.008

B STROBE Statement

STROBE Statement—Checklist of items that should be included in reports of ***cohort studies***

	Item No	Recommendation	Page No
Title and abstract	1	(*a*) Indicate the study's design with a commonly used term in the title or the abstract	Title
		(*b*) Provide in the abstract an informative and balanced summary of what was done and what was found	1
Introduction			
Background/rationale	2	Explain the scientific background and rationale for the investigation being reported	4
Objectives	3	State specific objectives, including any prespecified hypotheses	5
Methods			
Study design	4	Present key elements of study design early in the paper	6
Setting	5	Describe the setting, locations, and relevant dates, including periods of recruitment, exposure, follow-up, and data collection	6
Participants	6	(*a*) Give the eligibility criteria, and the sources and methods of selection of participants. Describe methods of follow-up	6
		(*b*) For matched studies, give matching criteria and number of exposed and unexposed	NA
Variables	7	Clearly define all outcomes, exposures, predictors, potential confounders, and effect modifiers. Give diagnostic criteria, if applicable	7
Data sources/ measurement	8*	For each variable of interest, give sources of data and details of methods of assessment (measurement). Describe comparability of assessment methods if there is more than one group	7
Bias	9	Describe any efforts to address potential sources of bias	NA
Study size	10	Explain how the study size was arrived at	6
Quantitative variables	11	Explain how quantitative variables were handled in the analyses. If applicable, describe which groupings were chosen and why	7
Statistical methods	12	(*a*) Describe all statistical methods, including those used to control for confounding	9
		(*b*) Describe any methods used to examine subgroups and interactions	NA
		(*c*) Explain how missing data were addressed	9
		(*d*) If applicable, explain how loss to follow-up was addressed	7
		(*e*) Describe any sensitivity analyses	9
Results			
Participants	13*	(a) Report numbers of individuals at each stage of study—eg numbers potentially eligible, examined for eligibility, confirmed eligible, included in the study, completing follow-up, and analysed	10
		(b) Give reasons for non-participation at each stage	10
		(c) Consider use of a flow diagram	Fig 1

Descriptive data	14*	(a) Give characteristics of study participants (eg demographic, clinical, social) and information on exposures and potential confounders	Tab 1
		(b) Indicate number of participants with missing data for each variable of interest	Tab 1
		(c) Summarise follow-up time (eg, average and total amount)	Tab 1
Outcome data	15*	Report numbers of outcome events or summary measures over time	Tab 2

Main results	16	(*a*) Give unadjusted estimates and, if applicable, confounder-adjusted estimates and their precision (eg, 95% confidence interval). Make clear which confounders were adjusted for and why they were included	10
		(*b*) Report category boundaries when continuous variables were categorized	NA
		(*c*) If relevant, consider translating estimates of relative risk into absolute risk for a meaningful time period	NA
Other analyses	17	Report other analyses done—eg analyses of subgroups and interactions, and sensitivity analyses	10
Discussion			
Key results	18	Summarise key results with reference to study objectives	12
Limitations	19	Discuss limitations of the study, taking into account sources of potential bias or imprecision. Discuss both direction and magnitude of any potential bias	16
Interpretation	20	Give a cautious overall interpretation of results considering objectives, limitations, multiplicity of analyses, results from similar studies, and other relevant evidence	10
Generalisability	21	Discuss the generalisability (external validity) of the study results	16
Other information			
Funding	22	Give the source of funding and the role of the funders for the present study and, if applicable, for the original study on which the present article is based	Title

*Give information separately for exposed and unexposed groups.

Note: An Explanation and Elaboration article discusses each checklist item and gives methodological background and published examples of transparent reporting. The STROBE checklist is best used in conjunction with this article (freely available on the Web sites of PLoS Medicine at http://www.plosmedicine.org/, Annals of Internal Medicine at http://www.annals.org/, and Epidemiology at http://www.epidem.com/). Information on the STROBE Initiative is available at http://www.strobe-statement.org.

C MBS Item numbers

Appendix Table C1. MBS item numbers, 85% benefit, and description used in this study.

Information reflects the MBS as of 1 July 2023.

Item	85% Benefit	Description
55028	100.05	* Head, ultrasound scan of (R)
55032	100.05	* Neck, one or more structures of, ultrasound scan of (R)
55036	101.95	* Abdomen, ultrasound scan of (including scan of urinary tract when performed), for morphological assessment, if: (a) the service is not solely a transrectal ultrasonic examination of any of the following:(i) prostate gland;(ii) bladder base;(iii) urethra; and(b) within 24 hours of the service, a service mentioned in item 55038 is not performed on the same patient by the providing practitioner (R)
55048	100.35	* Scrotum, ultrasound scan of (R)
55065	90.05	* Pelvis, ultrasound scan of, by any or all approaches, if:(a) the service is not solelya service to which an item (other thanitem 55736 or 55739) in Subgroup 5 of this Group applies ora transrectal ultrasonic examination of any of the following: prostate gland; bladder base; urethra; and (b) within 24 hours of the service, a service mentioned in item 55038 is not performed on the same patient by the providing practitioner (R)
55070	90.05	* Breast, one, ultrasound scan of (R)
55073	90.05	* Breast, one, ultrasound scan of (R)
55084	90.05	* Urinary bladder, ultrasound scan of, by any or all approaches, if within 24 hours of the service, a service mentioned in item 11917, 55036, 55038, 55065, 55600 or 55603 is not performed on the same patient by the providing practitioner (R)
55812	100.05	* Chest or abdominal wall, one or more areas, ultrasound scan of, if the service is not performed in conjunction with a service mentioned in item 55070, 55073, 55076 or 55079 (R)
55856	100.05	* Hand or wrist or both, left or right, ultrasound scan of, if the service is not performed in conjunction with a service mentioned in item 55858 (R)
55860	100.05	* Forearm or elbow, or both, left or right, ultrasound scan of, if the service is not performed in conjunction with a service mentioned in item 55862 (R)
55864	100.05	* Shoulder or upper arm, or both, left or right, ultrasound scan of, if:(a) the service is used for the assessment of one or more of the following suspected or known conditions:(i) an injury to a muscle, tendon or muscle/tendon junction; (ii) rotator cuff tear, calcification or tendinosis (biceps, subscapular, supraspinatus or infraspinatus);(iii) biceps subluxation;(iv) capsulitis and bursitis;(v) a mass, including a ganglion;(vi) an occult fracture;(vii) acromioclavicular joint pathology; and(b) the service is not performed in conjunction with a service mentioned in item 55866 (R)
55867	100.05	* Shoulder or upper arm, or both, left and right, ultrasound scan of, if:(a) the service is used for the assessment of one or more of the following suspected or known conditions:(i) an injury to a muscle, tendon or muscle/tendon junction;(ii) rotator cuff tear, calcification or tendinosis (biceps, subscapular, supraspinatus or infraspinatus);(iii) biceps subluxation;(iv) capsulitis and bursitis;(v) a mass, including a ganglion;(vi) an occult fracture;(vii) acromioclavicular joint pathology; and(b) the service is not performed in conjunction with a service mentioned in item 55865 (NR)
55868	100.05	* Hip or groin, or both, left or right, ultrasound scan of, if the service is not performed in conjunction with a service mentioned in item 55870 (R)
55876	100.05	* Buttock or thigh, or both, left or right, ultrasound scan of, if the service is not performed in conjunction with item 55878 (R)
55880	100.05	* Knee, left or right, ultrasound scan of, if: (a) the service is used for the assessment of one or more of the following suspected or known conditions:(i) abnormality of tendons or bursae about the knee;(ii) a meniscal cyst, popliteal fos

		sa cyst, mass or pseudomass;(iii) a nerve entrapment or a nerve or nerve sheat h tumour;(iv) an injury of collateral ligaments; and (b) the service is not perfo rmed in conjunction with item 55882 (R)
55884	100.05	* Lower leg, left or right, ultrasound scan of, if the service is not performed i n conjunction with item 55886 (R)
55886	111	* Lower leg, left and right, ultrasound scan of, if the service is not performed in conjunction with item 55884 (R)
55892	100.05	* Mid foot or fore foot, or both, left or right, ultrasound scan of, if the service is not performed in conjunction with item 55894 (R)
56007	229.1	* Computed tomography— scan of brain with intravenous contrast medium and with any scans of the brai n before intravenous contrast injection, when performed, not being a service t o which item 57007 applies (R) (Anaes.)
56028	308.6	* Computed tomography— scan of facial bones, para nasal sinuses or both with intravenous contrast med ium and with any scans of the facial bones, para nasal sinuses or both before i ntravenous contrast injection, when performed (R) (Anaes.)
56107	311.55	* Computed tomography— scan of soft tissues of neck, including larynx, pharynx, upper oesophagus and salivary glands (not associated with cervical spine)— with intravenous contrast medium and with any scans of soft tissues of neck, i ncluding larynx, pharynx, upper oesophagus and salivary glands (not associat ed with cervical spine) before intravenous contrast injection, when undertake n, not being a service associated with a service to which item 56807 applies (R) (Anaes.)
56219	298.95	Computed tomography— scan of spine, one or more regions with intrathecal contrast medium, includin g the preparation for intrathecal injection of contrast medium and any associat ed plain X rays, not being a service to which item 59724 or 59275 applies (R) (Anaes.)
56224	321.95	* Computed tomography— scan of spine, cervical region, with intravenous contrast medium and with any scans of the cervical region of the spine before intravenous contrast injection when undertaken (R) (Anaes.)
56225	321.95	* Computed tomography— scan of spine, thoracic region, with intravenous contrast medium and with an y scans of the thoracic region of the spine before intravenous contrast injectio n when undertaken (R) (Anaes.)
56226	321.95	* Computed tomography— scan of spine, lumbosacral region, with intravenous contrast medium and wit h any scans of the lumbosacral region of the spine prior to intravenous contra st injection when undertaken (R) (Anaes.)
56238	321.95	* Computed tomography— scan of spine, 3 regions, cervical, thoracic and lumbosacral, with intravenous contrast medium and with any scans of these regions of the spine before intra venous contrast injection when undertaken (R) (Anaes.)
56307	366.55	* Computed tomography— scan of chest, including lungs, mediastinum, chest wall and pleura, with or wi thout scans of the upper abdomen, with intravenous contrast medium and wit h any scans of the chest, including lungs, mediastinum, chest wall or pleura a nd upper abdomen before intravenous contrast injection, when undertaken, no t being a service to which item 56807 or 57007 applies and not including a st udy performed to exclude coronary artery calcification or image the coronary arteries (R) (Anaes.)
56407	329.9	* Computed tomography— scan of upper abdomen only (diaphragm to iliac crest), with intravenous contr ast medium, and with any scans of upper abdomen (diaphragm to iliac crest) before intravenous contrast injection, when undertaken, not being a service to which item 56307, 56507, 56807 or 57007 applies (R) (Anaes.)

56412	329.9	* Computed tomography— scan of pelvis only (iliac crest to pubic symphysis), with intravenous contrast medium and with any scans of pelvis (iliac crest to pubic symphysis) before i ntravenous contrast injection, when undertaken, not being a service to which i tem 56407 applies (R) (Anaes.)
56501	352.75	* Computed tomography— scan of pelvis only (iliac crest to pubic symphysis), with intravenous contrast medium and with any scans of pelvis (iliac crest to pubic symphysis) before i ntravenous contrast injection, when undertaken, not being a service to which i tem 56407 applies (R) (Anaes.)
56507	439.9	* Computed tomography— scan of upper abdomen and pelvis with intravenous contrast medium and with any scans of upper abdomen and pelvis before intravenous contrast injection, when performed, not for the purposes of virtual colonoscopy and not being a service to which item 56807 or 57007 applies (R) (Anaes.)
56623	306.65	* Computed tomography— scan of lower limb, left or right or both, one region (other than knee), or more than one region (which may include knee), with intravenous contrast mediu m and with any scans of the lower limb before intravenous contrast injection, when performed, not being a service to which item 56626 applies (R) (Anaes.) (Anaes.)
56627	201.6	* Computed tomography— scan of upper limb, left or right or both, any one region, or more than one regi on, without intravenous contrast medium (R) (Anaes.) (Anaes.)
56630	306.65	* Computed tomography— scan of upper limb and lower limb, left or right or both, any one region (other than knee), or more than one region (which may include knee) with intraven ous contrast medium with any scans of the limbs before intravenous contrast i njection, when performed, not being a service to which item 56626 applies (R) (Anaes.)
56807	513.15	* Computed tomography— scan of chest, abdomen and pelviswith or without scans of soft tissues of neck with intravenous contrast medium and with any scans of chest, abdomen and pelvis with or without scans of soft tissue of neck before intravenous contrast injection, when performed, not including a study performed to exclude coron ary artery calcification or image the coronary arteries (R) (Anaes.)
56807	513.15	* Computed tomography— scan of chest, abdomen and pelvis with or without scans of soft tissues of nec k with intravenous contrast medium and with any scans of chest, abdomen an d pelvis with or without scans of soft tissue of neck before intravenous contra st injection, when performed, not including a study performed to exclude coro nary artery calcification or image the coronary arteries (R) (Anaes.)
57007	520.2	* Computed tomography— scan of brain and chest with or without scans of upper abdomen with intraven ous contrast medium and with any scans of brain and chest and upper abdome n before intravenous contrast injection, when performed, not including a stud y performed to exclude coronary artery calcification or image the coronary art eries (R) (Anaes.)
57509	36.45	* Hand, wrist, forearm, elbow or humerus (R)
57521	39.8	* Foot, ankle, leg or femur (R)
57523	39.8	* Knee (R)
57703	49.5	* Shoulder or scapula (R)
57715	55.85	* Pelvic girdle (R)
57901	59.1	* Skull, not in association with item 57902 (R)
58100	61.5	* Spine—cervical (R)
58103	50.5	* Spine—thoracic (R)
58106	70.55	* Spine—lumbosacral (R)
58503	43.2	* Chest (lung fields) by direct radiography (R)

58903	43.65	* Plain abdominal only, not being a service associated with a service to which item 58909, 58912 or 58915 applies (R)
61421	407.85	* Bone study— whole body, with, when undertaken, blood flow, blood pool and delayed imaging on a separate occasion (R)
61505	85	* CT scan performed at the same time and covering the same body area as single photon emission tomography or positron emission tomography for the purpose of anatomic localisation or attenuation correction if no separate diagnostic CT report is issued and performed in association with a service to which an item in Subgroup 1 or 2 of Group I4 applies (R)
61553	905.8	* Whole body FDG PET study, following initial therapy, performed for the evaluation of suspected metastatic or recurrent malignant melanoma in patients considered suitable for active therapy (R)
63001	360.75	* MRI— scan of head (including MRA, if performed) for tumour of the brain or meninges (R) (Anaes.) (Contrast) (Anaes.)
63111	440.95	* MRI— scan of head and cervical spine (including MRA, if performed) for tumour of the central nervous system or meninges (R) (Anaes.) (Contrast) (Anaes.)
63154	320.7	* MRI— scan of one region or 2 contiguous regions of the spine for tumour (R) (Anaes.) (Contrast) (Anaes.)
63204	353.7	* MRI— scan of 3 contiguous or 2 non contiguous regions of the spine for tumour (R) (Anaes.) (Contrast) (Anaes.)
63301	300.65	* MRI— scan of musculoskeletal system for tumour arising in bone or musculoskeletal system, excluding tumours arising in breast, prostate or rectum (R) (Anaes.) (Contrast) (Anaes.)

D Imaging scan durations and carbon footprint

Estimates of scan duration used in this study, with assistance provided by Nicholas Marks of the Hunter New England Local Health District.

Computed tomography (CT) scan

10 min (1 site), 15 min (2 sites), 20 min (3 sites), 25 min (4 sites), 30 min (5 sites)

Positron emission tomography (PET) scan

90 min

PET and CT scan

100 min

Ultrasound

20 min (1 site), 40 min (2 sites), 60 min (3 sites)

X-ray

5 min (1 site), 10 min (2 sites)

Magnetic resonance imaging (MRI)

30 min (1 site), 60 min (2 sites), 90 min (3 sites)

Estimates of per-minute attributional carbon emissions were obtained from the existing life cycle assessment by McAlister et al. and are reported in the Table below.

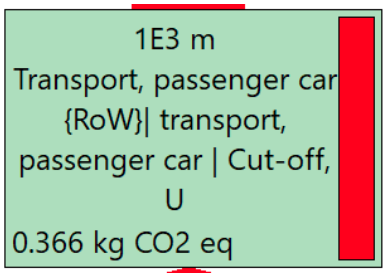
Appendix Table D-1: Per-minute attributional carbon emissions used in this study.

Scan type	**Per-minute CO_2-e emissions (kg)**
CT	1.15
Ultrasound	0.0265
X-Ray	0.38
MRI	0.6481
PET Scan	1.15
PET & CT	1.15

E AusLCI and Ecoinvent methods

***Transport by car* – for participants from NSW and ACT**

The AusLCI market process 'Transport, passenger car {RoW} | transport, passenger car | Cut-off, U' was used to calculate emissions for 1 km driven. This was 0.366 kg CO_2-e per km). This was then multiplied by the distance driven by each patient.



Appendix Figure E-1. Screenshot of output from SimaPro for carbon emissions from 1 km driven by car using AusLCI and Ecoinvent databases.

Transport by plane – for participants from QLD, TAS, VIC

The following AusLCI market processes were used to calculate carbon emissions for flights between Melbourne, Hobart, Brisbane and Sydney:

- Transport, passenger aircraft, short haul {GLO} | market for transport, passenger aircraft, very short haul | Cut-off, U
- Transport, passenger aircraft, very short haul {GLO} | market for transport, passenger aircraft, very short haul | Cut-off, U

The distance flown was calculated using an online calculator (https://www.airmilescalculator.com/distance/: accessed 27 March 2025).

Appendix Table E-1: Carbon emission associated with flights between Melbourne, Brisbane, Hobart and Sydney.

Origin	Destination	Distance	AusLCI process	CO2-e (one way)	CO2-e (return)
Melbourne	Sydney	706 km	Very short haul	121 kg	243 kg
Brisbane	Sydney	751 km	Very short haul	129 kg	258 kg
Hobart	Sydney	1037 km	Short haul	137 kg	273 kg

F Carbon valuation

In this study carbon values were obtained from the New South Wales Government's Carbon emissions in the Investment Framework Report. These carbon values were developed in alignment with the NSW Government Guide to Cost-Benefit Analysis with a 5% central discount rate.

Details are provided in the table below.

Appendix Table C1. Details of NSW Carbon Values.

Scenario	**Scenario description**	**Carbon value FY25 (AUD per tonne CO_2-e)**
High	Reflects lower volume of abatement, speed of deployment or cost-effectiveness of decarbonisation solutions, for example, due to supply chain/ workforce/ approval constraints.	230
Central	Fulfills NSW emissions reduction targets with orderly deployment of decarbonisation solutions. Central carbon values are the best translated of the optimised marginal abatement costs of NSW's decarbonisation process.	130
Low	Reflects higher volume of abatement, speed of deployment or cost-effectiveness of decarbonisation solutions, for example, due to the state shifting away from high-emission industries.	90

Source: NSW carbon emissions in the Investment Framework Report